COOLSHADE · DESIGN PRECIS

CoolShade design precis

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DOCUMENT CONTROL

Document ID	CSH-PRC-001	Version	0.7
Status	Draft	Date	2026-10-02
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/coolshade		
Source	docs/02-concept.md @ ae091ec		

REVISION HISTORY

Version	Date	Author	Change
0.7	2026-10-02	Amish Chadha	First site type and partner decided by Amish on 2026-10-02; first candidates named
0.6	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.5	2026-09-30	Amish Chadha	Constructable design (CSH-DDR-003) and build plan CSH-BLD-001; posts on a 2.8 x 2.35 m grid, panel at the rear edge, cleats and rivet nuts, cost $882 (R13 not met, new budget proposed)
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3 update from CSH-CAL-001 and CSH-DDR-001 (numbers checked, shorter knee braces, vacuum breaker, closed mist line loop, tank strap, design choices adopted for TRL 3, GA drawing CSH-DWG-001)
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, components, first-order numbers, safety, media)

0.1	2026-09-25	Amish Chadha	Initial scaffold

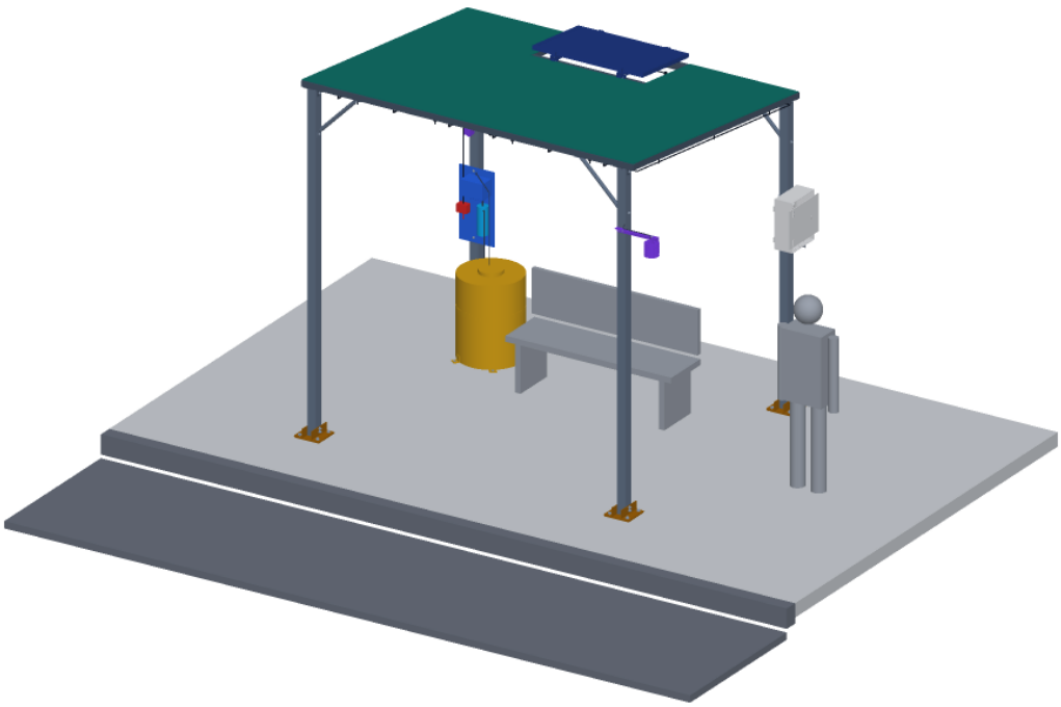
CoolShade design precis

Summary

CoolShade is a bolted steel shade canopy, 3.0 x 2.4 m in plan, with a knitted shade cloth roof, a 100 W solar panel, a 12 V battery and a low-pressure misting line. It gives shade all day and mists only when the air is hot and dry enough for misting to help and a person is waiting. The TRL 3 calculations (CSH-CAL-001) give, for a hot, dry design day (38 °C, 25 % relative humidity), 85 L of water and 203.8 Wh of electricity per day, an air temperature drop of 2.71 °C while spraying at 1 m/s wind (1.35 °C averaged over the cycle) and, from published shade sail data, a drop in mean radiant temperature of about 17 °C, which is where most of the relief comes from. On 2026-09-25 Amish accepted the recommendations (CSH-DDR-001 and CSH-DDR-002): R4 is now judged while spraying, R10 with the fabric removed above 15 m/s gusts, the battery is 25 Ah and the value-engineering target is $775. On 2026-09-30 the design was made constructable (CSH-DDR-003): every part is now modelled as it is made or bought and every joint as it is bolted, and the prototype build plan CSH-BLD-001 shows how to build it. The parts added for that bring the estimated cost to $882 against the $775 value-engineering target (a hypothetical control target), so R13 is over the target by $107 (savings worth trying are in CSH-DEC-001); water hygiene (R9) and nozzle scaling (R16) are at risk, and wetting (R5) needs droplet data.

CoolShade github.com/BoujeeEnjinia1701/coolshade

CONCEPT, NOT FOR FABRICATION



Seen from the front right and above, 24 deg elevation. Grey figure: 1.75 m person for scale

Figure 1. CoolShade at a curbside stop next to an existing bench, with a 1.75 m person for scale. Kit parts are colored; the street and bench are grey.

How it works

1. **Shade.** Four 80 mm square steel posts on a 2.8 x 2.35 m grid carry two horizontal long beams at the roof's street and rear edges; sloping end beams and purlins between them make a mono-pitch roof frame (7.1°, high edge at the street), with knitted HDPE shade cloth laced over it. The cloth blocks most direct sun while letting hot air escape.
2. **Harvest and store.** A 100 W panel on rails at the rear edge of the roof, in a notch in the cloth, charges a 12.8 V, 25 Ah LiFePO4 battery through an MPPT controller. Everything runs at 12 V DC.
3. **Sense.** A temperature and humidity sensor in a radiation shield on an arm off the front post measures the ambient air outside the mist zone. A passive infrared sensor under the front beam detects whether anyone is waiting. No camera or microphone is fitted.
4. **Decide.** The controller mists only when air temperature is 32 °C or more, relative humidity is 60 % or less, someone has been detected in the last 2 min, and the tank is above its low level (thresholds decided by Amish, CSH-DDR-001 D2). A 10 s control loop with a 20 s debounce stops the spray within 30 s of a condition failing. In humid air it does not mist, because misting would add little cooling and wet people.
5. **Mist.** A 12 V diaphragm pump draws from a 120 L opaque tank through a 5 µm filter and pushes water at about 7 bar through a closed loop of mist line just inboard of the long beams and under the end beams to eight anti-drip nozzles along the front and rear runs, in cycles of 20 s on and 20 s off. Fine droplets evaporate and cool the air under the canopy.
6. **Drain.** When the pump stops at the end of a session, a normally open drain valve at the low point empties the line in about 30 s, while a vacuum breaker at the front, highest point of the loop lets air in past the anti-drip nozzles. On design days the tank is drained of its residual and refilled with about 95 L daily, so no water is more than 24 h old, and it is disinfected weekly.

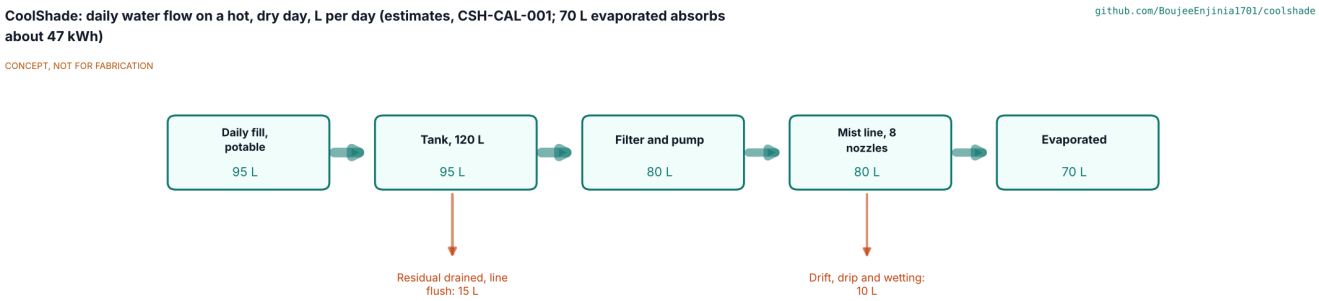


Figure 2. Daily water flow on the design day, in liters. All values are estimates.

Main components

Numbers match the exploded view (Figure 3) and `bom/bom.csv`.

Table 1. Main components.

#	Component	Choice	Notes
1	Posts with knee braces	80 x 80 x 3 mm galvanized SHS, 2.61 and 2.90 m, on a 2.8 x 2.35 m grid; one 32 mm tube brace each with flattened, bolted ends, bends 350 mm down and 350 mm along the beam	Utilization 0.30 in a 30 m/s gust (CSH-CAL-001)
2	Roof frame	50 x 50 mm SHS long beams on the post tops, end beams and a panel bearer; two 40 x 40 mm purlins; angle cleats with M8 bolts into rivet nuts	About 7° mono-pitch; no welding
3	Shade fabric	Knitted HDPE, about 90 % UV block, 3.0 x 2.4 m with a notch under the panel, laced over the frame	Fitted only in forecast gusts below 15 m/s (CSH-DDR-001, D6; CSH-DDR-002)
4	Base plates and anchors	220 x 220 x 10 mm plates and two 75 x 75 mm angle cleats per post, four M16 anchors each, into an existing slab	Footing checked per site
5	Solar panel	100 W monocrystalline on two rails and four L-feet, at the roof's rear edge	About 1,000 x 670 mm
6	Battery	12.8 V, 25 Ah LiFePO4 with BMS and cold-charge cutoff	Inside item 8, fused; 25 Ah for energy margin (CSH-DDR-002)
7	MPPT charge controller	12 V, 10 A, LiFePO4 profile	Inside item 8
8	Controller and enclosure	Lockable IP65 box on two rails on the rear right post, about 1.44 to 1.86 m high; microcontroller, pump and valve drivers, fuses	Firmware not started (TRL 4)
9	Sensor head	SHT4x-class temperature and humidity sensor in a multi-plate shield on an angle arm at 2.4 m, plus a PIR presence sensor	No camera or microphone
10	Misting pump	12 V diaphragm pump, about 7 bar, about 60 W, pressure switch	On an equipment plate on the rear left post, with items 11 and 14
11	Filter and check valve	5 µm cartridge, inlet strainer	Protects 0.4 mm nozzles
12	Mist line and nozzles	About 12 m of 9.5 mm (3/8 in) line as a closed loop on hanger clips, 8 brass anti-drip nozzles, about 0.4 mm	32 L/h while spraying at 4 L/h per nozzle (to confirm)
13	Water tank	120 L, opaque, food-grade, lockable, with a low-level switch; strapped to the rear left post, with two stop cleats	Optional mains float valve with backflow preventer; strap because an empty tank tips in a 30 m/s gust
14	Drain valve and vacuum breaker	12 V normally open solenoid at the low point; air admittance valve at the loop's high point	Empties the line in about 30 s after every session
15	Hose and wiring harness	Suction and delivery hose, outdoor cable, glands	

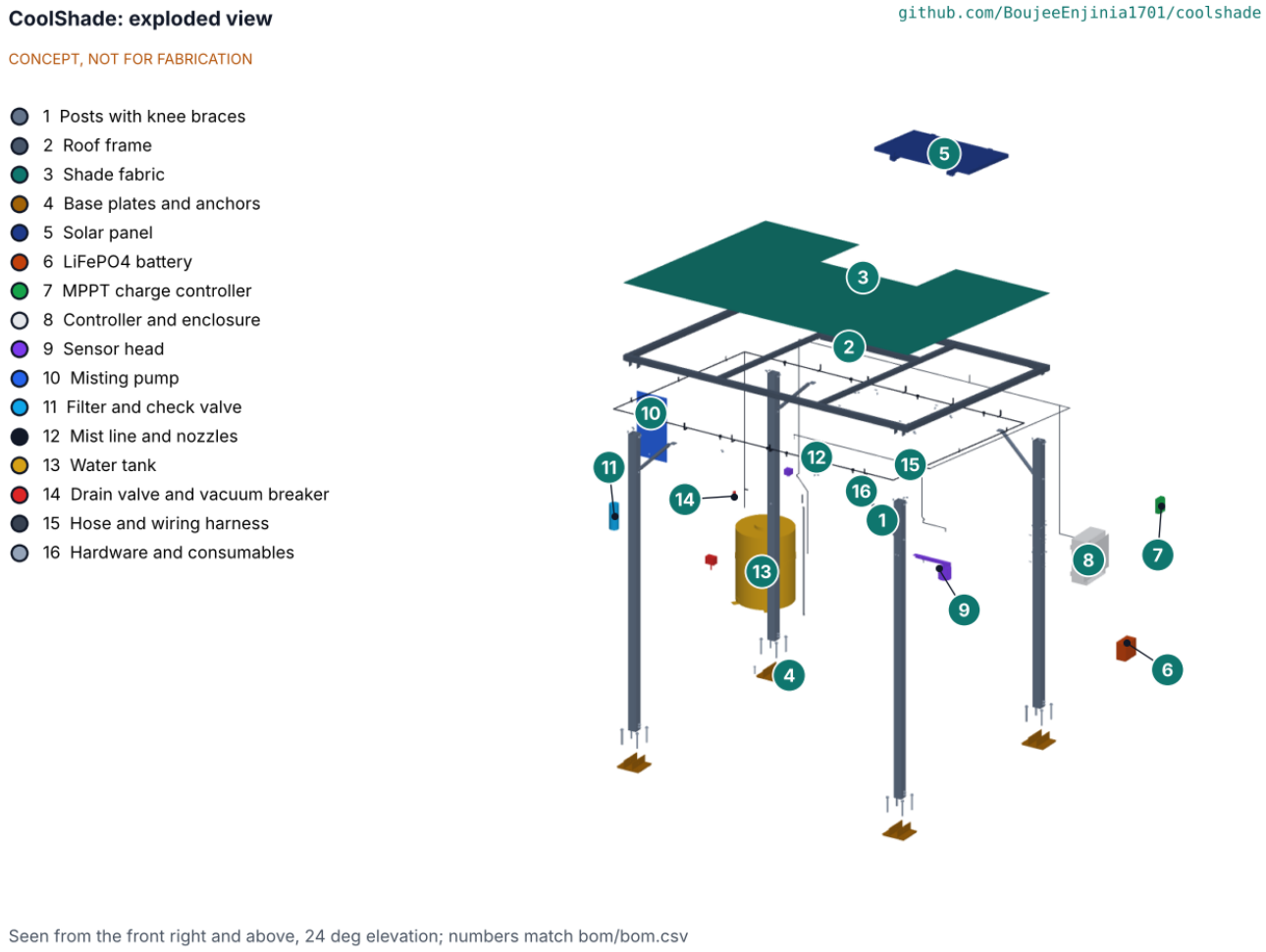


Figure 3. Exploded view with numbered callouts matching the BOM. Items 6 and 7 are shown out of the enclosure (item 8) that houses them.

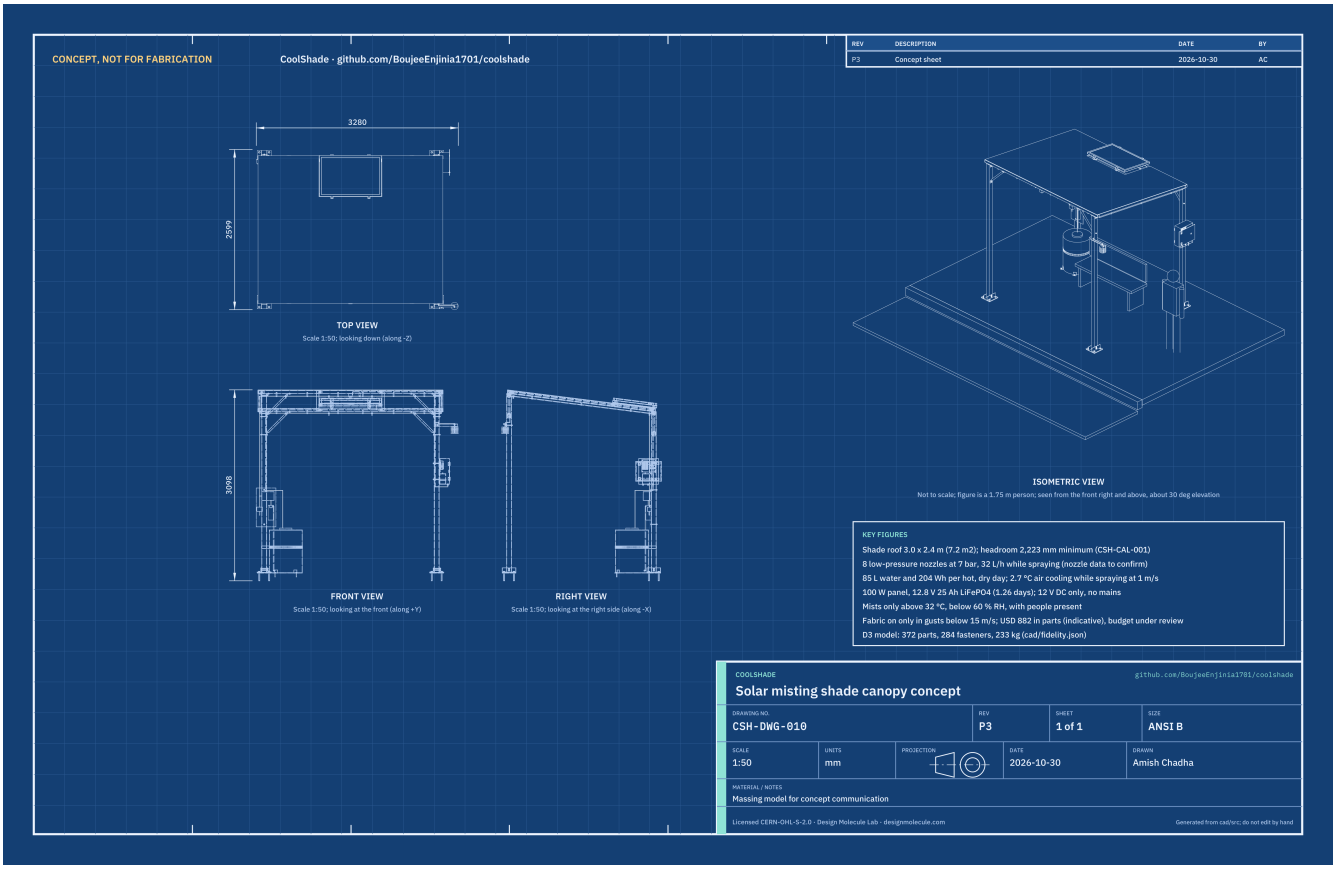


Figure 4. Concept sheet CSH-DWG-010 with orthographic and isometric views and key figures.

The general arrangement drawing CSH-DWG-001 (Rev P4, `cad/drawings/CSH-DWG-001.pdf`) gives the main dimensions and interfaces from the parametric model `cad/src/model.py`. It is preliminary and not for fabrication.

No cutaway is provided: the canopy is open, and the only enclosed parts (items 6 to 8) are shown in the exploded view.

Numbers checked at TRL 3

All values come from CSH-CAL-001 and its script `docs/04-calcs/sizing.py`; the tags in brackets are lines of its output. They are first-principles estimates, not measurements.

Water and misting

Assumptions: eight nozzles at 4 L/h each at 7 bar (to be checked against nozzle datasheets; this implies a discharge coefficient of 0.24 for a 0.4 mm orifice), 20 s on and 20 s off while active (50 % duty), 5 h of active misting on the design day, 87 % of sprayed water evaporating in the air, and a daily flush of 5 L.

Table 2. Water use on the design day.

Quantity	Value	Basis
Flow while spraying	32 L/h	8 x 4 L/h [C2]
Average flow while active	16 L/h	50 % duty
Sprayed per day	80 L	16 L/h x 5 h
Used per day, with flush	85 L	80 + 5 L
Drawn per day with the drain-and-refill routine	95 L	Daily fill; about 10 L residual drained [C5]
Evaporated in the air	69.6 L	87 % of 80 L
Tank endurance	1.41 design days	120 L / 85 L [C4]
Mist line volume and drain time	0.40 L, about 30 s	11.91 m of 6.5 mm bore; 3 mm valve orifice [C7]

R7 holds up to 4.75 L/h per nozzle [C3]. Whether the nozzles keep people dry (R5) depends on droplet size: droplets up to about 100 µm evaporate before reaching head height on the design day, but only up to about 80 µm in the most humid air the controller allows [D3]. No droplet data are in hand.

Cooling

Assumptions: latent heat of evaporation 2.43 MJ/kg; air density 1.15 kg/m³ and specific heat 1,006 J/(kg K) at 35 °C; air crossing a control volume 3.0 m long and 2.0 m high at the wind speed; all evaporation within that volume.

Table 3. Cooling on the design day.

Quantity	Value	Basis
Heat absorbed by evaporation, averaged	9.40 kW	13.9 L/h evaporated x 2.43 MJ/kg [E1]
Heat absorbed while spraying	18.8 kW	27.8 L/h evaporated
Air through the zone at 1 m/s	6.90 kg/s	6 m² x 1 m/s x 1.15 kg/m³
Air temperature drop at 1 m/s	2.71 °C while spraying, 1.35 °C averaged	Q / (m x cp) [E2]
Wind speed for a 2 °C drop while spraying (R4 as restated)	1.35 m/s or less	Same method [E3]
Wind speed for a 2 °C averaged drop (former R4)	0.68 m/s or less	Same method [E3]
Air temperature drop at 0.3 m/s	4.5 °C averaged	Wetting risk rises in still air [E3]
Wet-bulb temperature, 38 °C and 25 % RH	23.0 °C	Stull approximation; far from the limit [E4]
Humidity added at 1 m/s	0.56 g/kg	Small against about 10 g/kg ambient
Heat absorbed per day	47 kWh	69.6 L x 2.43 MJ/kg [E5]
Mean radiant temperature drop from shade	about 17 °C	Shade sails in Tempe (Middel et al., 2021)

The shade does most of the work: a 17 °C drop in mean radiant temperature matters more to a person standing in the sun than a 1 to 3 °C drop in air temperature. Misting adds a noticeable but smaller benefit, and only in dry air. This is why the controller must never mist in humid air. R4 is judged while spraying (CSH-DDR-002), where the 2.71 °C drop at 1 m/s meets the 2 °C target; averaged over the cycle it would need calmer air. No fans are fitted (CSH-DDR-001, D3, confirmed in CSH-DDR-002), because the shade gives most of the relief and fans would need a larger battery.

Energy

Table 4. Daily energy budget on the design day.

Load or source	Value	Basis
Pump	150 Wh	60 W x 50 % x 5 h [F1]
Controller and sensors	28.8 Wh	1.2 W x 24 h
Drain valve held closed while active	25 Wh	5 W x 5 h
Total use	**203.8 Wh**	
Panel harvest	337.5 Wh	100 W x 4.5 h x 0.75 [F2]
Battery usable	256.0 Wh	12.8 V 25 Ah, 320 Wh x 80 %; 1.256 design days

The panel covers the design day 1.66 times, and the 25 Ah battery covers 1.256 design days without sun, so R8 is met (the former 20 Ah battery gave 1.005 days, with 1 Wh to spare). A latching drain valve would save 24.5 Wh, but it would not open by itself if power failed, so the normally open valve is kept [F3]. The enclosure runs about 3.6 K above the air, so the battery stops charging (at 45 °C) when the air is above about 41.4 °C [F5].

Structure and wind

Assumptions: 30 m/s gust, air density 1.2 kg/m³, net normal force coefficient 1.2 on the roof with the fabric treated as solid, center of pressure a quarter of the depth off center, S275 steel with a partial factor of 1.5 on wind, fabric sag 5 % of its span.

Table 5. Wind check.

Quantity	Value	Basis
Dynamic pressure	540 Pa	0.5 x 1.2 x 30² [G2]
Normal force on the roof	4.67 kN	540 Pa x 1.2 x 7.2 m²
Post base moment	1.24 kN m	Horizontal roof load 1.21 kN plus post drag, cantilevers [G3]
Post stress, 80 x 80 x 3 SHS	54 MPa, utilization 0.30 factored	Z = 22,861 mm³ [G4]
Long beam, roof pressure as a simple span	Utilization 0.70	50 x 50 x 2 SHS over 2.8 m [G6]
Long beam, fabric edge pull at 5 % sag	3.89 kN/m, utilization 3.52	Membrane tension p L² / (8 f) [G7]
Fabric-on gust limit for the beams	about 16 m/s (58 km/h)	Edge pull the beams can take, 1.11 kN/m [G8]
Anchor tension, factored, original fabric-on case	about 6.4 kN per M16 anchor	Uplift plus base moment [G10]
Restated R10, fabric off at 30 m/s	Posts 0.14, panel bearer 0.02, purlins 0.07; anchors about 2.8 kN	Panel as a flat plate [G13]
Restated R10, fabric on at 15 m/s	Long beams 0.88, posts 0.07; anchors about 1.5 kN	Same solid-fabric case [G14]
Ballast needed if not anchored	309 kg	Uplift less dead load [G11]

The posts are adequate. The weak point is the fabric: a tensioned cloth pulls its edges inward far harder than the pressure alone suggests, and on the conservative solid-fabric case it overloads the 50 mm long beams in a 30 m/s gust. Porosity and stretch would reduce the pull, but no data are in hand. R10 is therefore restated (CSH-DDR-002): the frame and anchors survive 30 m/s with the fabric removed, and the fabric is fitted only when forecast gusts are below 15 m/s (54 km/h), where the beams reach 0.88. The restated R10 is met on paper. Taking the fabric off before storms (CSH-DDR-001, D6) is structurally necessary. The anchors must be checked against the chosen anchor's data and the actual slab. The unit must be anchored; ballast alone is impractical. An empty tank would tip over in the same gust, so it is strapped to the rear left post [G12].

Mass and cost

- Steel about 167 kg with its cleats and plates (a front post 21.0 kg, the heaviest part); panel about 7 kg; battery about 3 kg; full tank about 128 kg [G1, H1].
- Value-engineering target: $775 (`budget_usd` , a hypothetical control target, not a limit). Estimated cost of the constructable design: $882 (indicative, see `bom/bom.csv`), $107 over the target after the parts added to make the design buildable (CSH-DDR-003) [J1]. Savings worth trying are in CSH-DEC-001. The steel prices look low for small quantities [J2].

Key design choices

Decided by Amish, 2026-09-25: go with recommendation (CSH-DDR-001, CSH-DDR-002).

1. Low-pressure (about 7 bar) 12 V misting rather than high-pressure (about 70 bar) mains misting, so the unit runs off-grid and stays low-voltage. The trade-off is larger droplets and a higher wetting risk.
2. Humidity and presence gating with fixed thresholds, so water and energy are used only when misting helps someone.
3. Tank supply with an optional mains connection, so it can go anywhere.
4. Drain-after-use, a vacuum breaker and a daily drain-and-refill as the main defenses against *Legionella*.
5. Knitted shade cloth rather than a solid roof: cheaper, lighter and lets hot air escape, but it does not keep rain off, and it must come off before storms.
6. Bolted, movable frame on existing slabs, so a city can trial it at a site and move it.
7. No fans and no status link in the first prototype.

Integration with other lab projects

- **HeatMap Node** measures street-level heat stress and can show where a CoolShade would help most, and whether it did.
- **FieldNode** could provide an optional LoRaWAN status link (counts and levels only). Its 6 W panel is far too small to run the pump, so CoolShade keeps its own power system. No link is fitted in the first prototype (CSH-DDR-001, D7), so CoolShade places no interface demands on FieldNode.

Safety

Safety: Misting is an aerosol, and *Legionella* grows in water at 20 to 50 °C and infects people who inhale contaminated mist (WHO). Use potable water only; keep the tank opaque, shaded and locked; drain the line after every session; drain the tank's residual and refill it daily; disinfect weekly; and stop misting if the water is not maintained. A site owner must name a responsible person for water hygiene.

Safety: The 12.8 V LiFePO4 battery can overheat and burn if shorted, damaged or charged outside its limits. Fuse it at the terminal, use a BMS with cold-charge cutoff, keep it in the ventilated, locked enclosure, and never install a damaged or wet battery.

Safety: The pump line runs at about 7 bar. Depressurize before opening fittings, and wear eye protection when working on it.

Safety: The canopy is a wind-loaded structure. Anchor it to a checked slab or footing, strap the tank, fit the fabric only when forecast gusts are below 15 m/s (54 km/h) and remove it before storms (on the conservative calculation the roof beams are overloaded with the fabric on above gusts of about 16 m/s), and inspect bolts and anchors regularly. Building it is work at height with heavy steel; use two people, stable ladders and gloves, and deburr all cut edges.

Safety: Wet surfaces can be slippery. Set nozzles and duty cycle so people and the ground stay dry; stop misting and adjust if puddles form.

> **Safety:** CoolShade is shade and local cooling, not a medical intervention or a cooling center. People with heat illness need medical help.

Open questions

Open decisions and items to confirm are kept in the design decisions register (`docs/06-design-decisions.md`, CSH-DEC-001); the list below is the concept's own.

- [ ] Nozzle flow and droplet spectrum at 7 bar from the maker's data (R5, R7).
- [ ] Fabric porosity and stretch under wind, which decide the fabric's edge pull; a load-release cord lacing (TRL 4, on hold).
- [ ] Water hygiene plan acceptable to the local health authority, including sampling (R9).
- [ ] Hard water scaling and a descaling method (R16).
- [ ] Site anchor details on different slabs; footing option where no slab exists.
- [ ] Steel quotes, since the $882 estimated parts cost is over the $775 value-engineering target (R13).
- [x] First site type and co-design partner: decided 2026-10-02, a market lane with the market operator as co-design partner; first candidate to approach a farmers' market operator in the Phoenix area, with Arizona State University's urban heat researchers as the candidate measurement partner (CSH-DEC-001).